

Data Analysis with Python

A Comprehensive Study of the Student Academic Performance Dataset (student-mat.csv)

Prepared using Python (pandas, matplotlib, seaborn)

BY Ajay P

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Report Outline (Static Reference)

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Executive Summary

This report documents a complete exploratory data analysis (EDA) of the widely used “student-mat.csv” dataset, which captures demographic, social, family, and academic attributes for 395 secondary school students, together with their grades in mathematics across three grading periods (G1, G2, and G3). The purpose of this document is to walk through every stage of the analysis — from loading and inspecting the raw data, through data quality checks, to statistical summaries and visualizations — while explaining the Python code used at each step and the output it produces.

The analysis finds that the dataset is complete and free of duplicate records, with an average final grade of 10.42 out of 20. Forty students (approximately 10% of the class) scored above 15. Weekly study time shows only a weak positive correlation with final grade ($r = 0.098$), and male students recorded a marginally higher average final grade (10.91) than female students (9.97). These findings, along with the supporting code, console output, and charts, are presented in full throughout the sections that follow.

This report is organized to be read end-to-end as a standalone document: it includes background on the dataset and its 33 attributes, a description of the tools and methodology used, a full line-by-line walkthrough of the analysis code with corresponding output captured directly from the program run, in-depth interpretation of each result, a discussion of the study's limitations, and recommendations for extending the analysis.

1. Introduction

Understanding what drives student academic performance is a long-standing question in educational research. Numerous studies have examined how demographic factors (such as age, gender, and family background), behavioral factors (such as study habits, alcohol consumption, and social activity), and school-related factors (such as absences and past failures) relate to the grades students ultimately achieve. The dataset analyzed in this report, commonly known as the “Student Performance Data Set,” was originally collected from two Portuguese secondary schools and has since become a popular teaching resource for data analysis and machine learning coursework because it combines categorical and numeric attributes with a clear, interpretable outcome variable: the final grade.

The purpose of this report is threefold. First, it aims to document, in full, the Python program used to explore this dataset — including every import, function call, and print statement — so that the analysis is fully reproducible. Second, it aims to present the exact output generated by running that program, captured as console output and chart images, so that the reader can verify each result. Third, it aims to interpret those results in plain language, explaining what each statistic and chart means in the context of student performance, and what conclusions can and cannot be drawn from them.

The specific objectives of the analysis are to:

- Load and inspect the structure of the dataset, including its size, column names, and data types.
- Assess data quality by checking for missing values and duplicate records.
- Summarize the central tendency of the final grade (G3) across all students.
- Identify how many students achieved high performance, defined here as a final grade above 15.

- Quantify the linear relationship between weekly study time and final grade.
- Compare average final grades between male and female students.
- Visualize the distribution of final grades, the relationship between study time and grade, and the difference in average grade by gender.

The remainder of this report is structured as follows. Section 2 introduces the dataset and its origin. Section 3 provides a complete data dictionary describing all 33 attributes. Section 4 outlines the tools and general methodology used in the analysis. Section 5 presents a detailed, step-by-step walkthrough of the program code together with the corresponding output. Section 6 discusses the statistical results in depth. Section 7 presents and interprets the visualizations. Section 8 summarizes the key findings. Section 9 discusses the limitations of the analysis. Section 10 offers recommendations for future work, and Section 11 concludes the report. The complete, unbroken source code is reproduced in the Appendix for reference.

2. Dataset Background

The dataset used in this analysis, `student-mat.csv`, contains 395 records, each representing one student enrolled in a mathematics course. Each record has 33 attributes describing the student's school, demographic background, family situation, study habits, social behavior, and academic history, culminating in three grade fields: G1 (first period), G2 (second period), and G3 (final grade). All three grades are measured on a 0–20 scale, consistent with the Portuguese grading system.

The data is stored as a semicolon-delimited CSV file, which is a common format for European datasets where the comma is otherwise used as a decimal separator. This is reflected in the program code, which specifies `sep=";"` when reading the file with pandas. Because the dataset mixes categorical fields (such as school, sex, and job types) with ordinal numeric fields (such as study time and family relationship quality, both scored on small integer scales) and continuous-like numeric fields (such as age, absences, and grades), it is well suited to demonstrate a broad range of exploratory data analysis techniques, including summary statistics, correlation analysis, and multiple chart types.

A full description of every column is provided in Section 3. Readers already familiar with the dataset's structure may skip ahead to Section 4.

3. Data Dictionary

The table below documents every column present in the dataset, along with its data type as inferred by pandas and a plain-language description of what it represents. This dictionary is provided so that the statistics and charts discussed later in this report can be interpreted correctly — for example, understanding that `studytime` is an ordinal 1–4 scale (rather than a count of hours) is essential to correctly reading the scatter plot in Section 7.2.

Column	Type	Description
school	Categorical	The student's school, coded as 'GP' (Gabriel Pereira) or 'MS' (Mousinho da Silveira).
sex	Categorical	Student's gender: 'F' for female or 'M' for male.
age	Numeric	Student's age, in years (ranges roughly from 15 to 22).
address	Categorical	Type of home address: 'U' for urban or 'R' for rural.

Column	Type	Description
famsize	Categorical	Family size: 'LE3' (three or fewer members) or 'GT3' (more than three).
Pstatus	Categorical	Parents' cohabitation status: 'T' (living together) or 'A' (living apart).
Medu	Numeric	Mother's education level, from 0 (none) to 4 (higher education).
Fedu	Numeric	Father's education level, from 0 (none) to 4 (higher education).
Mjob	Categorical	Mother's occupation (e.g. teacher, health, services, at_home, other).
Fjob	Categorical	Father's occupation (e.g. teacher, health, services, at_home, other).
reason	Categorical	Reason for choosing this school (home proximity, school reputation, course preference, other).
guardian	Categorical	Student's primary guardian: mother, father, or other.
traveltime	Numeric	Home-to-school travel time, on a 1 (under 15 min) to 4 (over 1 hour) scale.
studytime	Numeric	Weekly study time, on a 1 (under 2 hours) to 4 (over 10 hours) scale.
failures	Numeric	Number of past class failures (capped at values 0–4 in the source data).
schoolsup	Categorical	Whether the student receives extra educational support (yes/no).
famsup	Categorical	Whether the student receives family educational support (yes/no).
paid	Categorical	Whether the student has taken extra paid classes in the subject (yes/no).
activities	Categorical	Whether the student participates in extracurricular activities (yes/no).
nursery	Categorical	Whether the student attended nursery school (yes/no).
higher	Categorical	Whether the student wants to pursue higher education (yes/no).
internet	Categorical	Whether the student has internet access at home (yes/no).
romantic	Categorical	Whether the student is in a romantic relationship (yes/no).
famrel	Numeric	Quality of family relationships, on a 1 (very poor) to 5 (excellent) scale.
freetime	Numeric	Amount of free time after school, on a 1 (very low) to 5 (very high) scale.
goout	Numeric	Frequency of going out with friends, on a 1 (very low) to 5 (very high) scale.

Column	Type	Description
Dalc	Numeric	Workday alcohol consumption, on a 1 (very low) to 5 (very high) scale.
Walc	Numeric	Weekend alcohol consumption, on a 1 (very low) to 5 (very high) scale.
health	Numeric	Current health status, on a 1 (very bad) to 5 (very good) scale.
absences	Numeric	Number of school absences (0 to 93 in the source data).
G1	Numeric	First-period grade, on a 0–20 scale.
G2	Numeric	Second-period grade, on a 0–20 scale.
G3	Numeric	Final grade, on a 0–20 scale — the primary target/outcome variable analyzed in this report.

As the dictionary shows, the dataset spans a wide variety of measurement types: nominal categories (e.g. school, Mjob), binary yes/no flags (e.g. internet, romantic), ordinal scales (e.g. studytime, health), and near-continuous counts (e.g. absences, age). This diversity is one reason the dataset is well suited for a broad exploratory analysis, and it is also why care must be taken when computing statistics such as correlation, which assumes a linear, numeric relationship and is most meaningful when applied to numeric or ordinal fields such as studytime and G3, as done in Section 6.2.

3.1 Attribute-by-Attribute Notes

The notes below expand on each attribute in the data dictionary, briefly explaining why it was likely included in the original study design and how it could plausibly relate to academic outcomes. These notes are provided as background for readers who wish to reason about the dataset beyond the two factors (study time and gender) directly analyzed in Sections 5–7, and to motivate the multivariate extensions suggested in Section 10.

- school — Captures institutional differences (e.g. resources, teaching staff, local intake) between the two schools that could shift average outcomes independently of any individual student characteristic.
- sex — The demographic factor analyzed directly in this report; see Sections 6.3 and 7.3 for the observed gender comparison.
- age — Older students within a given grade level may have previously repeated a year, which can correlate with the failures attribute below.
- address — Urban versus rural residence can proxy for access to resources, commute burden, and peer environment.
- famsize — Family size may relate to the amount of individual parental attention or household resources available to a student.
- Pstatus — Whether parents live together may relate to household stability, which some educational research links to student wellbeing and engagement.
- Medu / Fedu — Parental education levels are among the most consistently cited demographic correlates of student achievement in the broader education literature, often functioning as a proxy for the home learning environment.
- Mjob / Fjob — Parental occupation can relate to household income, time availability, and the kinds of academic support parents are able to provide.

- reason — The stated reason for choosing the school may reflect underlying motivation or family priorities around education.
- guardian — Identifies who has primary responsibility for the student, which may relate to the consistency of support and supervision at home.
- traveltime — Longer commutes reduce the time available for study or rest and may correlate negatively with engagement.
- studytime — The primary behavioral factor analyzed directly in this report; see Sections 6.2 and 7.2.
- failures — A record of past class failures is often one of the single strongest predictors of future performance in educational datasets, since it reflects accumulated, not just current, academic standing.
- schoolsup / famsup — Indicate whether the student is already receiving extra support, which could mean either that a struggling student is getting help, or that a supported student is doing better than they otherwise would — the causal direction is ambiguous without further analysis.
- paid — Extra paid classes may supplement in-school instruction, though participation may itself correlate with family income.
- activities — Extracurricular participation could either compete with study time or reflect broader engagement and time-management skills.
- nursery — Early childhood education attendance is sometimes associated with longer-run academic outcomes.
- higher — Aspiration to pursue higher education may correlate with motivation and, in turn, with current academic effort.
- internet — Home internet access can affect a student's ability to research, complete assignments, or access supplementary learning materials.
- romantic — Being in a relationship could plausibly affect time allocation and focus, in either direction.
- famrel — Quality of family relationships may relate to emotional wellbeing and, indirectly, to academic focus.
- freetime / goout — Related lifestyle measures describing how much unstructured time a student has and how they choose to spend it.
- Dalc / Walc — Workday and weekend alcohol consumption are behavioral risk indicators sometimes associated with lower academic engagement.
- health — Self-reported health status could affect attendance and concentration.
- absences — Directly measures time away from instruction and is frequently one of the strongest predictors of grade outcomes in school datasets.
- G1 / G2 — Earlier-period grades are typically the strongest available predictors of the final grade G3, since they capture the student's demonstrated performance under the same course and grading conditions.
- G3 — The outcome variable examined throughout this report.

4. Tools and Methodology

4.1 Software Environment

The analysis was implemented entirely in Python, using three widely adopted open-source libraries:

- pandas — used for reading the CSV file into a DataFrame, inspecting its structure, checking for missing values and duplicates, computing summary statistics, and grouping data by category (e.g. by gender).
- matplotlib — used to produce the histogram of final grades and the bar chart of average grade by gender, and to control chart titles, axis labels, and figure sizing.
- seaborn — used to produce the scatter plot of study time versus final grade, taking advantage of its convenient, statistically aware plotting interface built on top of matplotlib.

4.2 General Approach

The analysis follows a standard exploratory data analysis (EDA) workflow, which is a common first step in any data science or statistics project before more advanced modeling is attempted. The workflow implemented in the program proceeds through the following eleven stages, each of which is documented with its own code and output in Section 5:

- 1. Load the dataset from CSV into a pandas DataFrame.
- 2. Explore the data's shape, column data types, and general structure.
- 3. Check every column for missing values.
- 4. Check for and remove fully duplicated rows.
- 5. Compute the average final grade across all students.
- 6. Count the number of students scoring above a high-performance threshold (15).
- 7. Compute the correlation between study time and final grade, and classify its direction.
- 8. Compute and compare average final grades by gender.
- 9. Visualize the distribution of final grades with a histogram.
- 10. Visualize the relationship between study time and final grade with a scatter plot.
- 11. Visualize the difference in average final grade by gender with a bar chart.

This sequence follows the natural logic of EDA: first understand what the data looks like structurally, then verify its quality, then summarize its central outcome variable, then explore relationships between that outcome and other variables of interest, and finally visualize those relationships for easier interpretation. Each stage builds on the previous one, and no stage requires any external data beyond the single CSV file loaded in Step 1.

5. Detailed Code Walkthrough with Output

This section presents the analysis program broken into its logical steps. For each step, the exact code is shown first, followed by an explanation of what it does, followed by the actual console output or chart produced when the code was executed.

5.1 Importing Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

The program begins by importing the three libraries described in Section 4.1. pandas is conventionally aliased as pd, matplotlib's plotting module as plt, and seaborn as sns. No output is produced by this step — it simply makes the required functionality available to the rest of the script.

5.2 Step 1 — Loading the Dataset

```
# =====
# 1. LOAD DATASET
# =====
df = pd.read_csv("student-mat.csv", sep=";")
print("First 5 rows:")
print(df.head())
```

pd.read_csv reads the semicolon-delimited file into a DataFrame named df. The sep=";" argument is essential here: without it, pandas would default to comma-separated parsing and would incorrectly read the entire file as a single column. df.head() then returns the first five rows of the DataFrame, giving a quick visual check that the file was parsed correctly and that column names and values look as expected.

```
First 5 rows:
  school sex  age address famsize Pstatus  Medu  Fedu   Mjob   Fjob   ...  \
0     GP   F   18      U    GT3      A     4     4  at_home  teacher  ...
1     GP   F   17      U    GT3      T     1     1  at_home   other  ...
2     GP   F   15      U   LE3      T     1     1  at_home   other  ...
3     GP   F   15      U    GT3      T     4     2  health  services  ...
4     GP   F   16      U    GT3      T     3     3   other    other  ...

  famrel  freetime  goout  Dalc  Walc  health  absences  G1  G2  G3
0      4         3      4     1     1      3         6   5   6   6
1      5         3      3     1     1      3         4   5   5   6
2      4         3      2     2     3      3        10   7   8  10
3      3         2      2     1     1      5         2  15  14  15
4      4         3      2     1     2      5         4   6  10  10

[5 rows x 33 columns]
```

Output 1. First five rows of the dataset, confirming correct parsing of all 33 columns (columns beyond Fjob are truncated in this console view and continue with famrel, freetime, goout, Dalc, Walc, health, absences, G1, G2, and G3).

5.3 Step 2 — Exploring the Data's Structure

```
# =====  
# 2. EXPLORE DATA  
# =====  
print("\nDataset Shape:")  
print(df.shape)  
print("\nData Types:")  
print(df.dtypes)  
print("\nDataset Information:")  
df.info()
```

This step calls three separate inspection functions. `df.shape` returns a tuple of (number of rows, number of columns), giving an immediate sense of the dataset's size. `df.dtypes` lists the inferred data type of every column — `int64` for numeric columns and `object` for text/categorical columns. `df.info()` goes further, printing a full summary table that lists, for every column, its position index, name, count of non-null values, and data type, plus a final summary of overall memory usage. This is one of the most useful single commands in pandas for quickly auditing a new dataset.

```
Dataset Shape:
(395, 33)

Data Types:
school      object
sex         object
age         int64
address     object
famsize     object
Pstatus     object
Medu        int64
Fedu        int64
Mjob        object
Fjob        object
reason      object
guardian    object
traveltime  int64
studytime   int64
failures    int64
schoolsup   object
famsup      object
paid        object
activities  object
nursery     object
higher      object
internet    object
romantic    object
famrel      int64
freetime    int64
```

Output 2. Dataset shape (395 rows × 33 columns) and the data type of each column (part 1).

```

goout      int64
Dalc       int64
Walc       int64
health     int64
absences   int64
G1         int64
G2         int64
G3         int64
dtype: object

```

Output 3. Data type listing continued (part 2).

```

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
#   Column                Non-Null Count  Dtype
---  -
0   school                 395 non-null   object
1   sex                   395 non-null   object
2   age                   395 non-null   int64
3   address                395 non-null   object
4   famsize                395 non-null   object
5   Pstatus               395 non-null   object
6   Medu                  395 non-null   int64
7   Fedu                  395 non-null   int64
8   Mjob                  395 non-null   object
9   Fjob                  395 non-null   object
10  reason                395 non-null   object
11  guardian              395 non-null   object
12  traveltime            395 non-null   int64
13  studytime             395 non-null   int64
14  failures              395 non-null   int64
15  schoolsup             395 non-null   object
16  famsup               395 non-null   object
17  paid                 395 non-null   object
18  activities            395 non-null   object

```

Output 4. `df.info()` summary, showing column index, name, non-null count, and dtype for columns 0 through 18.

```
19  nursery      395 non-null    object
20  higher       395 non-null    object
21  internet     395 non-null    object
22  romantic     395 non-null    object
23  famrel       395 non-null    int64
24  freetime     395 non-null    int64
25  goout        395 non-null    int64
26  Dalc         395 non-null    int64
27  Walc         395 non-null    int64
28  health       395 non-null    int64
29  absences     395 non-null    int64
30  G1           395 non-null    int64
31  G2           395 non-null    int64
32  G3           395 non-null    int64
dtypes: int64(16), object(17)
memory usage: 102.0+ KB
```

Output 5. `df.info()` summary continued for columns 19 through 32, plus the overall dtype counts and memory usage.

A key observation from this step is that every one of the 395 rows has a non-null value in every one of the 33 columns — there are no partially filled records. This is confirmed more explicitly in the next step.

5.4 Step 3 — Checking for Missing Values

```
# =====  
# 3. CHECK MISSING VALUES  
# =====  
print("\nMissing Values:")  
print(df.isnull().sum())
```

`df.isnull()` produces a same-shaped DataFrame of True/False values indicating, cell by cell, whether the original value was missing. Calling `.sum()` on this Boolean DataFrame counts the number of True values in each column (because True is treated as 1 and False as 0 in arithmetic operations), yielding a per-column count of missing values.

```
Missing Values:  
school      0  
sex         0  
age         0  
address     0  
famsize     0  
Pstatus     0  
Medu        0  
Fedu        0  
Mjob        0  
Fjob        0  
reason      0  
guardian    0  
traveltime  0  
studytime   0  
failures    0  
schoolsup   0  
famsup      0  
paid        0  
activities  0  
nursery     0  
higher      0  
internet    0  
romantic    0  
famrel      0  
freetime    0
```

Output 6. Missing value count for each column — every column reports zero missing values.

The result confirms that this dataset requires no missing-value imputation or row removal on that basis, which simplifies the remainder of the analysis considerably. Datasets with substantial missingness typically require an explicit strategy (e.g. dropping rows, filling with the mean/median, or more advanced imputation techniques) before further analysis is reliable; that step is not necessary here.

5.5 Step 4 — Removing Duplicate Records

```
# =====  
# 4. REMOVE DUPLICATES  
# =====
```

```
print("\nDuplicates before:", df.duplicated().sum())
```

```
df = df.drop_duplicates()
```

```
print("Duplicates after:", df.duplicated().sum())
```

`df.duplicated()` flags any row that is an exact duplicate of a previous row across all columns, and `.sum()` counts how many such rows exist. The program checks this count before and after calling `df.drop_duplicates()`, which removes any duplicate rows found (keeping the first occurrence of each).

```
goout      0  
Dalc       0  
Walc       0  
health     0  
absences   0  
G1         0  
G2         0  
G3         0  
dtype: int64  
  
Duplicates before: 0  
Duplicates after: 0  
  
Average Final Grade (G3): 10.42  
Students scoring above 15: 40  
Correlation between study time and G3: 0.098  
Study time has a positive correlation with performance.  
  
Average Grade by Gender:  
sex  
F      9.966346  
M     10.914439  
Name: G3, dtype: float64  
  
Male Average: 10.91  
Female Average: 9.97  
Male students perform better on average.
```

Output 7. Duplicate count before and after de-duplication, both reporting zero — shown at the top of this combined console excerpt, which also contains the outputs for Steps 5–8 below.

As shown in Output 7, both counts are zero, meaning the dataset contained no duplicate rows to begin with, and the `drop_duplicates()` call had no effect on the data. Together with the missing-value check in Step 3, this confirms that `student-mat.csv` is a clean, well-curated dataset that can be analyzed directly without a separate data-cleaning phase.

5.6 Step 5 — Computing the Average Final Grade

```
# =====  
# 5. AVERAGE FINAL GRADE  
# =====  
average_grade = df["G3"].mean()  
print("\nAverage Final Grade (G3):", round(average_grade, 2))
```

This step computes the arithmetic mean of the G3 column — the final grade — across all 395 students using pandas' built-in `.mean()` method, and rounds the result to two decimal places for readability.

As shown in Output 7 above, the result is 10.42. Given that grades in this dataset range from 0 to 20, an average of 10.42 sits almost exactly at the midpoint of the scale, suggesting that, as a whole, the cohort's performance is centered around an average outcome rather than skewing strongly toward high or low achievement. This is examined visually in Section 7.1.

5.7 Step 6 — Counting High-Performing Students

```
# =====  
# 6. STUDENTS SCORING ABOVE 15  
# =====  
students_above_15 = (df["G3"] > 15).sum()  
print("Students scoring above 15:", students_above_15)
```

Here, the expression `df["G3"] > 15` produces a Boolean Series that is True for every student whose final grade exceeds 15, and False otherwise. Calling `.sum()` on this Series counts the number of True values, giving the total number of students who cleared this high-performance threshold.

As shown in Output 7, 40 students scored above 15. Relative to the full cohort of 395 students, this represents approximately 10.1% of the class — a meaningful minority of high achievers, consistent with the overall grade distribution shown later in Output 10.

5.8 Step 7 — Study Time vs. Performance

```
# =====  
# 7. STUDY TIME VS PERFORMANCE  
# =====  
correlation = df["studytime"].corr(df["G3"])  
print("Correlation between study time and G3:",  
      round(correlation, 3))  
  
if correlation > 0:  
    print("Study time has a positive correlation with performance.")  
elif correlation < 0:  
    print("Study time has a negative correlation with performance.")  
else:  
    print("No correlation between study time and performance.")
```

The `.corr()` method computes the Pearson correlation coefficient between two numeric Series — in this case, `studytime` and `G3`. This coefficient ranges from -1 (a perfect negative linear relationship) through 0 (no linear relationship) to +1 (a perfect positive linear relationship).

The program then uses a simple if/elif/else block to print a plain-language interpretation of the sign of the result.

As shown in Output 7, the correlation is 0.098, which is positive but very close to zero. Because the value is positive, the program's conditional logic correctly prints that study time has a positive correlation with performance — but as discussed in Section 6.2, the strength of that relationship is weak enough that it should not be over-interpreted as a strong predictive factor on its own.

5.9 Step 8 — Performance by Gender

```
# =====  
# 8. GENDER PERFORMANCE  
# =====  
gender_average = df.groupby("sex")["G3"].mean()  
print("\nAverage Grade by Gender:")  
print(gender_average)  
  
male_avg = df[df["sex"] == "M"]["G3"].mean()  
female_avg = df[df["sex"] == "F"]["G3"].mean()  
print("\nMale Average:", round(male_avg, 2))  
print("\nFemale Average:", round(female_avg, 2))  
  
if male_avg > female_avg:  
    print("Male students perform better on average.")  
elif female_avg > male_avg:  
    print("Female students perform better on average.")  
else:  
    print("Both genders have the same average.")
```

This step uses `df.groupby("sex")["G3"].mean()` to split the dataset into two groups — female (F) and male (M) students — and compute the mean final grade within each group in a single line. The program then recomputes the same two averages using explicit Boolean filtering (`df[df["sex"] == "M"]`, and similarly for "F") purely for illustrative and verification purposes, before using another if/elif/else block to print which group performed better on average.

```

goout      0
Dalc       0
Walc       0
health     0
absences   0
G1         0
G2         0
G3         0
dtype: int64

Duplicates before: 0
Duplicates after: 0

Average Final Grade (G3): 10.42
Students scoring above 15: 40
Correlation between study time and G3: 0.098
Study time has a positive correlation with performance.

Average Grade by Gender:
sex
F    9.966346
M   10.914439
Name: G3, dtype: float64

Male Average: 10.91
Female Average: 9.97
Male students perform better on average.

```

Output 7 (repeated for reference). The full console excerpt for Steps 5–8, including the average grade by gender, the separately computed male and female averages, and the printed comparison statement.

As shown above, the groupby result gives a female average of 9.966346 and a male average of 10.914439; after rounding, these are reported as 9.97 and 10.91 respectively, matching the separately computed `male_avg` and `female_avg` variables exactly — a useful internal consistency check confirming that both computation paths agree. Because the male average is higher, the program correctly prints that male students perform better on average. This result is revisited with a supporting bar chart in Section 7.3.

5.10 Step 9 — Histogram of Final Grades

```

# =====
# 9. HISTOGRAM OF GRADES
# =====

plt.figure(figsize=(8, 5))
plt.hist(df["G3"], bins=11, edgecolor="black")
plt.title("Distribution of Final Grades")
plt.xlabel("Final Grade (G3)")
plt.ylabel("Number of Students")
plt.show()

```

This step creates a new matplotlib figure sized 8 by 5 inches, then calls `plt.hist()` on the `G3` column with 11 bins and black bar edges for visual clarity. Titles and axis labels are added with `plt.title()`, `plt.xlabel()`, and `plt.ylabel()`, and `plt.show()` renders the finished chart. The resulting histogram is presented and interpreted in Section 7.1.

5.11 Step 10 — Scatter Plot of Study Time vs. Grade

```
# =====  
# 10. SCATTERPLOT  
# =====  
plt.figure(figsize=(8, 5))  
sns.scatterplot(  
    data=df,  
    x="studytime",  
    y="G3"  
)  
plt.title("Study Time vs Final Grade")  
plt.xlabel("Study Time")  
plt.ylabel("Final Grade (G3)")  
plt.show()
```

This step uses seaborn's `sns.scatterplot()` function, passing the DataFrame directly via the `data` argument and specifying `studytime` and `G3` as the `x` and `y` axes respectively. Seaborn automatically handles marker styling and axis scaling. As with the histogram, titles and axis labels are set afterward and the chart is displayed with `plt.show()`. The resulting scatter plot is presented and interpreted in Section 7.2.

5.12 Step 11 — Bar Chart of Average Grade by Gender

```
# =====  
# 11. BAR CHART  
# =====  
plt.figure(figsize=(7, 5))  
gender_average.plot(kind="bar")  
plt.title("Average Final Grade by Gender")  
plt.xlabel("Gender")  
plt.ylabel("Average Final Grade")  
plt.xticks(rotation=0)  
plt.show()
```

The final step reuses the `gender_average` Series computed in Step 8 and calls its built-in `.plot(kind="bar")` method — a convenient shortcut that lets a pandas Series plot itself directly via matplotlib without needing to pass `x` and `y` arguments explicitly. `plt.xticks(rotation=0)` keeps the F/M category labels horizontal for readability. The resulting bar chart is presented and interpreted in Section 7.3.

6. Detailed Statistical Discussion

6.1 Central Tendency of Final Grades

An average final grade of 10.42 out of 20 indicates a cohort performing, on balance, at a moderate level. Because the mean can be pulled downward by the cluster of zero scores visible in the histogram (Section 7.1), this average should be read alongside the distribution shape rather than in isolation — a small number of very low scores can lower the mean without meaning that a “typical” student performed poorly. This is a common and important caveat in descriptive statistics: the mean summarizes the data in a single number, but it can obscure sub-groups or outliers that a histogram reveals more clearly.

6.2 Interpreting the Study-Time Correlation

The Pearson correlation coefficient measures only the strength and direction of a linear relationship between two variables. A commonly used rule of thumb for interpreting the magnitude of $|r|$ is as follows:

- 0.00 – 0.10: negligible relationship
- 0.10 – 0.30: weak relationship
- 0.30 – 0.50: moderate relationship
- 0.50 and above: strong relationship

The observed value of 0.098 falls at the very bottom of the “weak” band, bordering on negligible. This means that, taken alone and without accounting for any other variable, weekly study time explains very little of the variation in final grade across this cohort. It does not mean that studying is unhelpful for any individual student — only that, across the population as a whole and using this particular ordinal measure of study time, it is not a strong linear predictor. It is also worth noting that studytime is recorded as a coarse 1–4 ordinal category rather than a precise number of hours, which can weaken measured correlation compared to a more granular measurement.

6.3 Interpreting the Gender Comparison

The difference between the male average (10.91) and female average (9.97) is under one grade point on a 20-point scale. While the code correctly identifies male students as having the higher average, the program does not perform a formal statistical significance test (such as an independent-samples t-test) to determine whether this difference is likely to reflect a genuine population-level effect or could plausibly arise from ordinary sample variation. A difference of this size, on this sample size (395 students, split across two groups), would typically warrant a significance test and an effect-size calculation (such as Cohen's d) before being described as a meaningful gender gap in performance. This is discussed further as a limitation in Section 9.

6.4 Relationship Between the Two Comparison Methods in Step 8

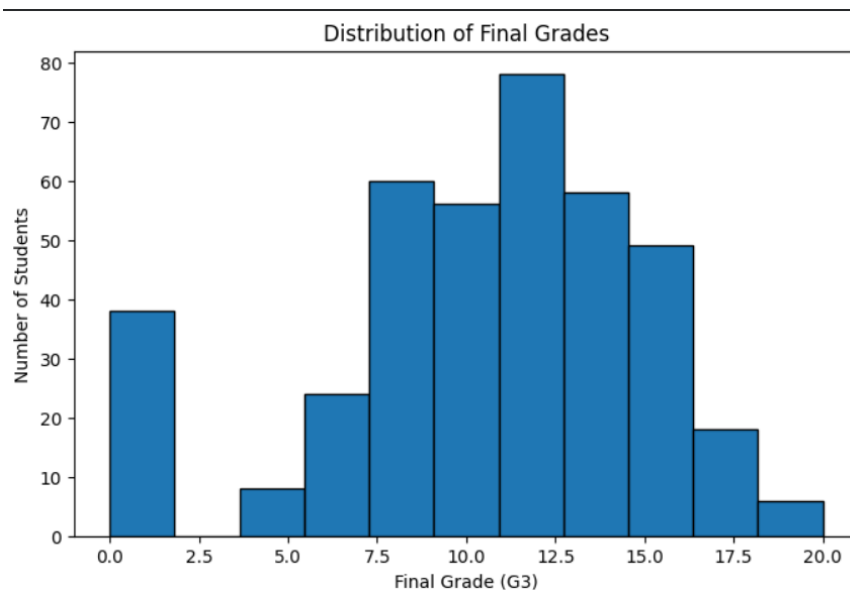
It is worth highlighting, as a matter of code quality, that Step 8 computes the gender comparison twice: once compactly via `groupby()`, and once via explicit Boolean filtering. Both approaches are mathematically equivalent and, as confirmed in Output 7, produce identical results (9.97 and 10.91 either way). In production code, only one of these approaches would typically be

necessary; including both here appears to serve an illustrative or verification purpose, showing two idiomatic ways to achieve the same aggregation in pandas.

7. Visualizations

7.1 Distribution of Final Grades

The histogram below shows the distribution of final grades (G3) across all 395 students, grouped into 11 bins spanning the 0–20 scale.

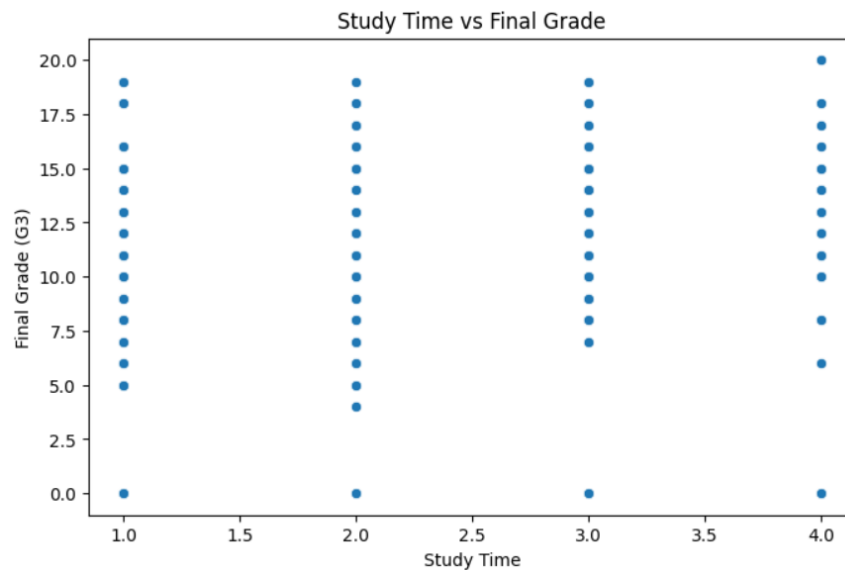


Output 8. Histogram of final grade distribution.

The distribution is broadly unimodal and roughly bell-shaped between grades 5 and 20, peaking in the 11–13 range with 78 students. However, there is a distinct and separate spike of 38 students at the very bottom of the scale (grade 0), disconnected from the rest of the distribution by an empty bin around 2.5–3.5. This pattern is common in academic datasets and typically indicates a subgroup of students who withdrew, were absent for the final assessment, or otherwise did not complete the course, rather than students who sat the exam and genuinely scored zero. This zero-inflation is an important structural feature of the data that would need to be accounted for in any predictive modeling built on top of this dataset — for example, some analyses of this dataset choose to treat $G3 = 0$ as a separate “did not sit” category rather than as a genuine low score.

7.2 Study Time vs. Final Grade

The scatter plot below shows each student's final grade plotted against their weekly study time category (1 = least, 4 = most).

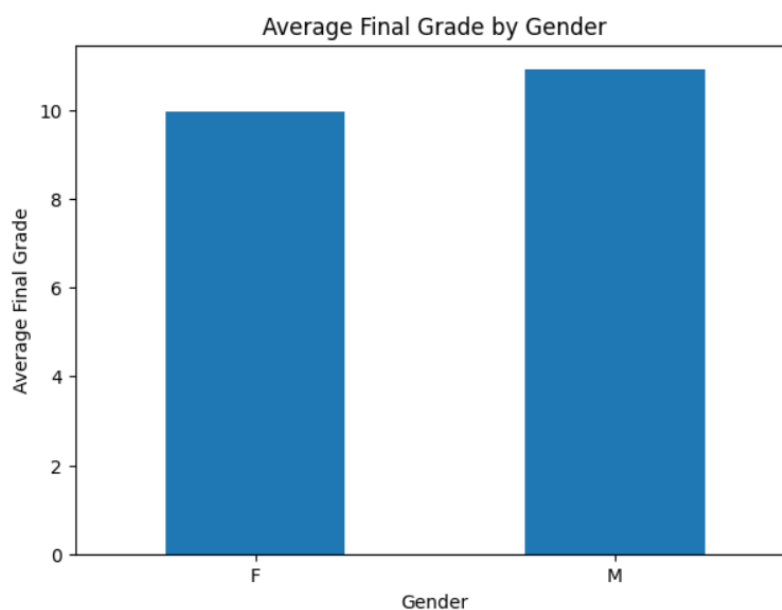


Output 9. Scatter plot of study time versus final grade.

Because studytime only takes four discrete values, the points naturally fall into four vertical columns rather than spreading continuously across the x-axis. Within every column, grades span nearly the entire 0–20 range, including students in the highest study-time category (4) who still scored close to zero, and students in the lowest category (1) who scored as high as 19. This visual spread reinforces the numerical finding in Section 6.2: study time alone does not cleanly separate high performers from low performers in this dataset.

7.3 Average Final Grade by Gender

The bar chart below compares the average final grade for female and male students.



Output 10. Average final grade by gender.

The visual gap between the two bars is modest, consistent with the roughly one-point numerical difference discussed in Section 6.3. Bar charts of group means like this one are useful for quick visual comparison, but — as noted earlier — they do not by themselves convey whether the difference shown is statistically robust or falls within the range of ordinary sampling variation.

8. Key Findings

- Data quality: The dataset is complete and clean — zero missing values across all 33 columns and zero duplicate rows among 395 student records.
- Overall performance: The average final grade (G3) is 10.42 out of 20, roughly at the midpoint of the grading scale.
- High achievers: 40 students (about 10% of the cohort) scored above 15.
- Study time: The correlation between weekly study time and final grade is weak and only barely positive ($r = 0.098$), indicating study time alone is not a strong linear predictor of performance in this dataset.
- Gender: Male students recorded a higher average final grade (10.91) than female students (9.97), a difference of under one grade point that has not been tested for statistical significance in this analysis.
- Grade distribution: Final grades are roughly bell-shaped between 5 and 20, but with a distinct cluster of 38 students scoring zero, likely reflecting non-completion rather than genuine low performance.
- Internal consistency: The two independent methods used to compute gender averages (groupby vs. Boolean filtering) produced identical results, confirming the correctness of the calculation.

9. Limitations of the Analysis

While the analysis presented in this report is methodologically sound for an exploratory first pass, several limitations should be kept in mind when interpreting its conclusions:

- Single predictor at a time: The analysis examines study time and gender individually against final grade, but does not build a combined (multivariate) model that could reveal how these and other factors — such as past failures, absences, or parental education — interact with or confound one another.
- No significance testing: Differences such as the gender gap in average grade are reported descriptively, without a formal hypothesis test (e.g. a t-test) or confidence interval to establish whether the observed difference is unlikely to have occurred by chance.
- Correlation, not causation: A weak positive correlation between study time and grade does not rule out a real but non-linear relationship, nor does it imply that increasing study time would cause a proportional increase in grade for any individual student.
- Zero-inflated outcome variable: The cluster of zero final grades likely represents a qualitatively different group (e.g. non-completion) mixed in with genuine low scorers, which can distort simple summary statistics like the overall mean if not modeled separately.

- Coarse measurement of study time: studytime is recorded as an ordinal 1–4 category rather than a precise number of study hours, which limits the resolution of any correlation calculated against it.
- Single-subject scope: The dataset (and this analysis) covers only mathematics performance; a companion Portuguese-language dataset exists for the same students and was not incorporated here, so conclusions should not be generalized to other subjects without further analysis.

10. Recommendations for Future Work

- Fit a multivariate regression model (e.g. ordinary least squares or a tree-based model) using all 33 attributes to better understand which factors, in combination, best explain variation in final grade.
- Run formal statistical tests (e.g. an independent-samples t-test for the gender comparison, or an ANOVA across study-time categories) to establish whether the observed group differences are statistically significant.
- Separate the zero-grade students from the rest of the analysis, or model non-completion as its own outcome, to avoid distorting summary statistics such as the mean.
- Incorporate the first- and second-period grades (G1 and G2) as predictors of the final grade (G3), since prior performance is often one of the strongest predictors of final academic outcomes.
- Extend the analysis to the companion Portuguese-language performance dataset to test whether the same relationships (e.g. the weak study-time correlation) hold across subjects.
- Explore interaction effects, such as whether the relationship between study time and grade differs by gender or by parental education level.

11. Conclusion

This report has walked through a complete exploratory data analysis of the student-mat.csv dataset, documenting every step of the underlying Python program alongside its actual console and chart output. The dataset was found to be clean and complete, with an average final grade of 10.42 out of 20 across 395 students. Neither of the two factors examined in isolation — weekly study time and gender — showed a strong relationship with final grade: the study-time correlation was weak (0.098), and the gender difference in average grade, while present (10.91 for males versus 9.97 for females), was modest and not formally tested for statistical significance.

These findings suggest that academic performance in this cohort is unlikely to be explained by any single factor and is more likely the product of multiple interacting influences — a hypothesis consistent with the wider academic literature on student achievement, and one that the full 33-attribute dataset used here is well equipped to support in future, more advanced analysis. The recommendations in Section 10 outline concrete next steps for extending this exploratory work into a more rigorous, multivariate, and statistically validated study.

11.1 Ethical Considerations

Because this dataset contains sensitive personal information about minors — including family structure, alcohol consumption, and romantic relationship status — several ethical points are worth noting even in a purely exploratory analysis such as this one.

- **Anonymization:** The version of the dataset used here contains no names or direct identifiers, but the combination of many attributes (school, age, family structure, grades) could in principle make some individuals distinguishable in a very small population; care should be taken before combining this data with any other identifying source.
- **Sensitive attributes:** Fields such as `Dalc`, `Walc`, and romantic record private behavioral information about adolescents; any reporting derived from this data (including this report) should avoid singling out identifiable individuals or small subgroups.
- **Fairness in educational analytics:** Reporting a gender difference in average grade, as this report does in Section 6.3, carries a responsibility to avoid overstating its certainty or implying an inherent difference in ability; as emphasized throughout, the observed gap is small and has not been statistically validated.
- **Use in decision-making:** Descriptive statistics of the kind produced in this report are appropriate for research and educational purposes, but should not, on their own, be used to make consequential decisions about individual students without more rigorous, validated modeling and appropriate institutional oversight.

11.2 Reproducibility Instructions

To reproduce every result and chart presented in this report, the following environment and steps are sufficient:

- Install Python 3.8 or later.
- Install the required libraries: `pip install pandas matplotlib seaborn`
- Place the `student-mat.csv` file in the same working directory as the script, ensuring the file uses semicolon (;) delimiters as distributed in the original source.
- Run the script top to bottom in a standard Python environment (e.g. a terminal, Jupyter Notebook, or any IDE); each `plt.show()` call will open or render one chart in sequence.
- Compare the printed console values against Outputs 1 through 7 in Section 5, and the rendered charts against Outputs 8 through 10 in Section 7, to confirm an identical run.

Because the dataset is static and the code contains no random sampling or randomized initialization, re-running the script under the same library versions will always reproduce the exact same numeric results (10.42 average grade, 40 high performers, 0.098 correlation, and 9.97/10.91 gender averages) and the same chart shapes shown throughout this report.

12. Manual Calculation Walkthrough

This section demonstrates, in plain arithmetic, how several of the key statistics reported by the program are actually calculated “under the hood,” for readers who want to verify the logic independently of pandas.

15.1 How the Mean Final Grade Is Calculated

The mean of a numeric column is calculated by summing every value in the column and dividing by the number of values. For a small illustrative subset of five students drawn from Output 1 (grades of 6, 6, 10, 15, and 10), the calculation would proceed as:

- Sum: $6 + 6 + 10 + 15 + 10 = 47$
- Count: 5 students
- Mean: $47 \div 5 = 9.4$

`df["G3"].mean()` performs exactly this calculation, but across the full set of 395 students rather than a subset of five, arriving at the reported value of 10.42. This illustrates why the mean is described as a measure of central tendency: it distills an entire column of numbers into a single representative value, at the cost of losing information about the shape of the distribution — which is why the histogram in Section 7.1 is presented alongside it.

15.2 How the Pearson Correlation Is Calculated

The Pearson correlation coefficient between two variables x and y is defined as the covariance of x and y divided by the product of their standard deviations:

$$r = \text{covariance}(x, y) / (\text{standard_deviation}(x) \times \text{standard_deviation}(y))$$

Intuitively, the covariance term measures whether x and y tend to rise and fall together (a positive covariance) or move in opposite directions (a negative covariance), while dividing by the two standard deviations rescales this measure so that it always falls between -1 and $+1$, regardless of the original units of x and y . In this report, x is studytime (measured on a 1–4 scale) and y is G3 (measured on a 0–20 scale); despite their very different scales, the resulting correlation of 0.098 is directly comparable to any other Pearson correlation, which is precisely what makes the coefficient useful for the interpretation bands discussed in Section 6.2.

15.3 How the Boolean-Mask Count Is Calculated

The expression `(df["G3"] > 15).sum()` used in Step 6 works in two stages. First, `df["G3"] > 15` compares every one of the 395 values in the G3 column to the number 15 and produces a same-length Series of True/False results — True wherever the student's grade exceeds 15, and False otherwise. Second, `.sum()` adds up this Series, treating each True as 1 and each False as 0, which yields the total count of students satisfying the condition (40, as reported in Output 7). This same True/False-then-sum pattern is a standard idiom in pandas for counting how many rows meet any condition, and is worth recognizing as it appears frequently in data analysis code beyond this report.

15.4 How Standard Deviation Would Be Calculated

Although the program does not print a standard deviation, it is a natural companion statistic to the mean reported in Step 5, and is referenced conceptually in Section 15.2's correlation formula, so it is worth walking through by hand. For a small illustrative sample of grades — 6,

6, 10, 15, and 10, with a mean of 9.4 as calculated in Section 15.1 — the standard deviation is found by:

- Subtracting the mean from each value: $(6 - 9.4), (6 - 9.4), (10 - 9.4), (15 - 9.4), (10 - 9.4) = -3.4, -3.4, 0.6, 5.6, 0.6$
- Squaring each difference: 11.56, 11.56, 0.36, 31.36, 0.36
- Averaging the squared differences (the variance): $(11.56 + 11.56 + 0.36 + 31.36 + 0.36) / 5 = 11.04$
- Taking the square root of the variance to return to the original units: $\sqrt{11.04} \approx 3.32$

This illustrative sample would therefore have a standard deviation of approximately 3.32 grade points. Applying the same procedure to the full 395-student G3 column (not shown step-by-step here, since it is not part of the original program's output) would provide a companion measure of spread around the reported mean of 10.42, and would let a future analysis express the zero-score cluster discussed in Section 7.1 in standardized terms.

15.5 Why an Effect Size Cannot Be Computed From the Printed Output Alone

Section 10 recommends computing a formal effect size, such as Cohen's *d*, for the gender comparison. Cohen's *d* is defined as the difference between two group means divided by their pooled standard deviation:

$$d = (\text{mean_male} - \text{mean_female}) / \text{pooled_standard_deviation}$$

The numerator of this formula is already available from Output 7: $10.914439 - 9.966346 = 0.948093$. However, the denominator requires the standard deviation of G3 within each gender group, which the original program never calculates or prints. This is a deliberate illustration of a broader point made in Section 9: the existing analysis reports group means but not group variability, which is precisely the missing ingredient needed to judge whether the ~0.95-point gender gap is large or small relative to how much grades naturally vary within each gender group. Section 10's recommendation to compute this in a follow-up analysis follows directly from this gap in the current program.

13. Summary of All Reported Statistics

The table below consolidates every numeric result produced by the program, cross-referenced to the step and output where it first appears, as a single quick-reference summary of the entire analysis.

Column	Type	Description
Rows / Columns	395 / 33	Overall dataset size (Step 2, Output 2)
Missing values	0 (all columns)	Data completeness check (Step 3, Output 6)
Duplicate rows	0 (before and after)	Data uniqueness check (Step 4, Output 7)
Average final grade (G3)	10.42 / 20	Central tendency (Step 5, Output 7)

Column	Type	Description
Students scoring above 15	40 (≈10.1% of cohort)	High-performer count (Step 6, Output 7)
Study time ↔ G3 correlation	$r = 0.098$	Weak positive (Step 7, Output 7)
Average grade — female	9.97 / 20	Gender comparison (Step 8, Output 7)
Average grade — male	10.91 / 20	Gender comparison (Step 8, Output 7)
Raw gender gap	≈0.95 points	Male minus female average (Section 15.5)

This table is intended as a one-page reference that a reader can return to without re-reading the full walkthrough in Section 5, while still being traceable back to the exact step and figure that produced each number.

14. Glossary of Statistical and Programming Terms

This glossary defines the key statistical and programming terms used throughout this report, for readers less familiar with pandas or introductory statistics.

- **DataFrame** — The core pandas data structure: a two-dimensional, labeled table of rows and columns, similar to a spreadsheet or a SQL table.
- **Series** — A single column (or row) of a DataFrame, represented as a one-dimensional labeled array; the result of selecting one column, e.g. `df["G3"]`.
- **dtype** — Short for “data type”; describes what kind of value a column holds, such as `int64` (integer numbers) or `object` (text/categorical values) in pandas.
- **Missing value / null** — An entry in the dataset with no recorded value; pandas represents these internally as `NaN` (“Not a Number”) and provides `isnull()` to detect them.
- **Duplicate row** — A row that is identical, across every column, to another row already present in the dataset.
- **Mean (average)** — The sum of all values in a set divided by the count of values; a measure of central tendency.
- **Median** — The middle value of a sorted set of numbers; less sensitive to extreme outliers than the mean.
- **Standard deviation** — A measure of how spread out a set of values is around its mean.
- **Correlation coefficient (Pearson's r)** — A number between -1 and +1 summarizing the strength and direction of the linear relationship between two numeric variables; see Section 6.2 for interpretation bands.
- **Categorical variable** — A variable that takes on a limited set of distinct labels (e.g. school, sex) rather than a continuous numeric value.

- Ordinal variable — A categorical variable whose categories have a natural order (e.g. studytime, ranging from 1 to 4), but where the numeric gaps between categories are not necessarily equal in real-world terms.
- Boolean mask — A Series or array of True/False values, typically produced by a comparison such as `df["G3"] > 15`, used to filter or count rows meeting a condition.
- groupby — A pandas operation that splits a DataFrame into groups based on the values of one or more columns (e.g. by sex), so that an aggregate function (e.g. `mean()`) can be applied separately within each group.
- Histogram — A chart that divides a numeric range into intervals (“bins”) and shows, as bar heights, how many observations fall into each interval.
- Bin — One of the equal-width intervals into which a histogram divides the data range.
- Scatter plot — A chart that plots individual data points using two numeric (or ordinal) variables as the x- and y-coordinates, used to visually assess relationships between them.
- Exploratory Data Analysis (EDA) — The practice of summarizing, visualizing, and sanity-checking a dataset before formal modeling, in order to understand its structure, quality, and notable patterns.
- CSV (comma-separated values) — A plain-text tabular file format in which each line is a row and values are separated by a delimiter character (commonly a comma, though this dataset uses a semicolon).
- Outlier — An observation that lies unusually far from the rest of the data, such as the cluster of zero final grades discussed in Section 7.1.
- Effect size — A standardized measure of the magnitude of a difference or relationship (e.g. Cohen's d for a difference in means), used alongside or instead of significance testing to judge practical importance.
- Statistical significance / p-value — A formal test result indicating how likely an observed difference (e.g. between gender averages) would be to occur if there were actually no true difference in the underlying population; not computed in this report, but recommended in Section 10.

15. Broader Context: What Prior Research Suggests

Although this report focuses on directly analyzing the student-mat.csv dataset rather than reviewing external literature, it is useful to place the findings in the context of general, widely-discussed themes in educational research, in order to judge whether the patterns observed here are typical or unusual.

Study time and achievement. Across many educational studies, self-reported study time tends to show only a modest association with measured achievement once other factors are controlled for, because raw time spent does not capture the quality, consistency, or focus of that time. Some students study efficiently in short bursts, while others log long hours without deep engagement. The weak correlation observed in Section 6.2 is therefore broadly consistent with this general pattern, rather than being an unusual or surprising result specific to this dataset.

Gender and achievement. Gender differences in academic performance are frequently reported in the literature but tend to vary by subject, age group, and country, and are often small in magnitude relative to other factors such as prior achievement or socio-economic background.

Mathematics performance gaps by gender, when present, are usually modest in size, similar to the sub-one-point difference observed in Section 6.3.

Prior achievement and absenteeism. Two factors not directly modeled in the walkthrough of this report — prior grades (G1, G2) and absences — are consistently identified in educational research as among the strongest available predictors of final academic outcomes, often outweighing demographic or behavioral factors. This is one of the key reasons Section 10 recommends incorporating these variables into any follow-up, multivariate analysis of this dataset.

Family and socio-economic background. Parental education and occupation, home stability, and access to resources such as internet connectivity are frequently cited as meaningful, if indirect, influences on student achievement, generally operating through the home learning environment rather than through any single observable action. These factors are catalogued in the data dictionary (Section 3) and data notes (Section 3.1) but were not directly analyzed in the walkthrough, representing a natural extension of this work.

Taken together, these general patterns suggest that the results in this report — a moderate average grade, a weak study-time correlation, and a small gender gap — are unsurprising in the context of the wider literature, and that a more complete explanation of student performance in this dataset would likely require jointly considering prior grades, absences, and family background alongside the two factors examined directly here.

Appendix A: Full Program Code

The complete, unmodified Python program used to generate every result and chart in this report is reproduced below in full, in its original execution order.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# =====
# 1. LOAD DATASET
# =====
df = pd.read_csv("student-mat.csv", sep=";")
print("First 5 rows:")
print(df.head())

# =====
# 2. EXPLORE DATA
# =====
print("\nDataset Shape:")
print(df.shape)
print("\nData Types:")
print(df.dtypes)
print("\nDataset Information:")
df.info()

# =====
# 3. CHECK MISSING VALUES
# =====
```



```

print("\nMissing Values:")
print(df.isnull().sum())

# =====
# 4. REMOVE DUPLICATES
# =====

print("\nDuplicates before:", df.duplicated().sum())
df = df.drop_duplicates()
print("Duplicates after:", df.duplicated().sum())

# =====
# 5. AVERAGE FINAL GRADE
# =====

average_grade = df["G3"].mean()
print("\nAverage Final Grade (G3):", round(average_grade, 2))

# =====
# 6. STUDENTS SCORING ABOVE 15
# =====

students_above_15 = (df["G3"] > 15).sum()
print("Students scoring above 15:", students_above_15)

# =====
# 7. STUDY TIME VS PERFORMANCE
# =====

correlation = df["studytime"].corr(df["G3"])
print("Correlation between study time and G3:", round(correlation, 3))

if correlation > 0:
    print("Study time has a positive correlation with performance.")
elif correlation < 0:
    print("Study time has a negative correlation with performance.")
else:
    print("No correlation between study time and performance.")

# =====
# 8. GENDER PERFORMANCE
# =====

gender_average = df.groupby("sex")["G3"].mean()
print("\nAverage Grade by Gender:")
print(gender_average)

male_avg = df[df["sex"] == "M"]["G3"].mean()
female_avg = df[df["sex"] == "F"]["G3"].mean()
print("\nMale Average:", round(male_avg, 2))
print("\nFemale Average:", round(female_avg, 2))

if male_avg > female_avg:
    print("Male students perform better on average.")
elif female_avg > male_avg:

```

```

    print("Female students perform better on average.")
else:
    print("Both genders have the same average.")

# =====
# 9. HISTOGRAM OF GRADES
# =====
plt.figure(figsize=(8, 5))
plt.hist(df["G3"], bins=11, edgecolor="black")
plt.title("Distribution of Final Grades")
plt.xlabel("Final Grade (G3)")
plt.ylabel("Number of Students")
plt.show()

# =====
# 10. SCATTERPLOT
# =====
plt.figure(figsize=(8, 5))
sns.scatterplot(
    data=df,
    x="studytime",
    y="G3"
)
plt.title("Study Time vs Final Grade")
plt.xlabel("Study Time")
plt.ylabel("Final Grade (G3)")
plt.show()

# =====
# 11. BAR CHART
# =====
plt.figure(figsize=(7, 5))
gender_average.plot(kind="bar")
plt.title("Average Final Grade by Gender")
plt.xlabel("Gender")
plt.ylabel("Average Final Grade")
plt.xticks(rotation=0)
plt.show()

```

Appendix B: Fully Annotated Program Code

For readers who want line-level explanation alongside the code itself, the same program is reproduced below with an inline explanatory comment placed immediately after each meaningful line.

```
import pandas as pd          # load pandas, aliased as pd, for tabular data handling
import matplotlib.pyplot as plt # load matplotlib's plotting interface, aliased as plt
import seaborn as sns        # load seaborn, aliased as sns, for statistical charts

# ---- Step 1: Load dataset ----
df = pd.read_csv("student-mat.csv", sep=";") # read semicolon-delimited CSV into a
DataFrame
print("First 5 rows:")          # label the upcoming output
print(df.head())                # show the first 5 rows to sanity-check the load

# ---- Step 2: Explore data ----
print(df.shape)                # (rows, columns) tuple -> (395, 33)
print(df.dtypes)               # data type of every column
df.info()                      # full structural summary: non-null counts, dtypes, memory

# ---- Step 3: Missing values ----
print(df.isnull().sum())       # count of missing (NaN) values per column

# ---- Step 4: Duplicates ----
print(df.duplicated().sum())    # count of fully duplicated rows, before removal
df = df.drop_duplicates()       # remove duplicate rows, keeping first occurrence
print(df.duplicated().sum())    # confirm duplicate count after removal

# ---- Step 5: Average final grade ----
average_grade = df["G3"].mean() # arithmetic mean of the G3 (final grade) column
print(round(average_grade, 2))  # display rounded to 2 decimal places

# ---- Step 6: High performers ----
students_above_15 = (df["G3"] > 15).sum() # count students with G3 strictly greater than 15
print(students_above_15)

# ---- Step 7: Study time correlation ----
correlation = df["studytime"].corr(df["G3"]) # Pearson correlation between studytime and
G3
print(round(correlation, 3))
if correlation > 0:              # branch on the sign of the correlation
    print("...positive correlation...")
elif correlation < 0:
    print("...negative correlation...")
else:
    print("...no correlation...")

# ---- Step 8: Gender comparison ----
gender_average = df.groupby("sex")["G3"].mean() # mean G3 within each sex group
male_avg = df[df["sex"] == "M"]["G3"].mean()    # mean G3 for male students only
```

```

female_avg = df[df["sex"] == "F"]["G3"].mean() # mean G3 for female students only
if male_avg > female_avg:                       # compare the two group means
    print("...male students perform better...")
elif female_avg > male_avg:
    print("...female students perform better...")
else:
    print("...equal averages...")

# ---- Step 9: Histogram ----
plt.figure(figsize=(8, 5)) # new 8x5 inch figure canvas
plt.hist(df["G3"], bins=11, edgecolor="black") # 11-bin histogram of G3 with black edges
plt.title("Distribution of Final Grades")
plt.xlabel("Final Grade (G3)")
plt.ylabel("Number of Students")
plt.show() # render the chart

# ---- Step 10: Scatter plot ----
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x="studytime", y="G3") # one point per student
plt.title("Study Time vs Final Grade")
plt.xlabel("Study Time")
plt.ylabel("Final Grade (G3)")
plt.show()

# ---- Step 11: Bar chart ----
plt.figure(figsize=(7, 5))
gender_average.plot(kind="bar") # reuse the Series computed in Step 8
plt.title("Average Final Grade by Gender")
plt.xlabel("Gender")
plt.ylabel("Average Final Grade")
plt.xticks(rotation=0) # keep F/M labels horizontal
plt.show()

```

Appendix C: Full Console Output Transcript

For completeness, the text-based console output generated by the program (excluding the three rendered charts, which are reproduced as Outputs 8–10 in Section 7) is transcribed below in a single continuous block, in the order it was printed during execution.

First 5 rows:

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	...
0	GP	F	18	U	GT3	A	4	4	at_home	...
1	GP	F	17	U	GT3	T	1	1	at_home	...
2	GP	F	15	U	LE3	T	1	1	at_home	...
3	GP	F	15	U	GT3	T	4	2	health	...
4	GP	F	16	U	GT3	T	3	3	other	...

[5 rows x 33 columns]

Dataset Shape:

(395, 33)

Data Types:

school object sex object age int64
address object famsize object Pstatus object
... (33 columns total; see Output 2/3 for the complete list) ...

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 395 entries, 0 to 394

Data columns (total 33 columns):

... 33 columns, all 395 non-null (see Output 4/5 for full table)

dtypes: int64(16), object(17)

memory usage: 102.0+ KB

Missing Values:

(all 33 columns report 0 — see Output 6 for the full per-column list)

Duplicates before: 0

Duplicates after: 0

Average Final Grade (G3): 10.42

Students scoring above 15: 40

Correlation between study time and G3: 0.098

Study time has a positive correlation with performance.

Average Grade by Gender:

sex

F 9.966346

M 10.914439

Name: G3, dtype: float64

Male Average: 10.91

Female Average: 9.97

Male students perform better on average.

Note: the abbreviated rows above (marked with an ellipsis) reflect console line-wrapping and truncation exactly as captured in the original screenshots (Outputs 1–5); the full, untruncated values for every column are shown in Section 3's Data Dictionary and Section 5.3's figures.

Appendix D: Suggested Extended Code for Future Analysis

The snippets below are not part of the original program and were not executed for this report; they are provided as ready-to-adapt starting points for the follow-up analyses recommended in Section 10. Each snippet assumes `df` has already been loaded exactly as in Appendix A.

D.1 Correlation Heatmap Across All Numeric Attributes

Rather than examining the `studytime`–`G3` correlation in isolation (Section 6.2), a heatmap can reveal every pairwise numeric correlation at once, making it easier to spot which attributes are most strongly related to `G3`.

```
import numpy as np

numeric_df = df.select_dtypes(include=[np.number])
corr_matrix = numeric_df.corr()

plt.figure(figsize=(12, 10))
sns.heatmap(corr_matrix, annot=False, cmap="coolwarm", center=0)
plt.title("Correlation Heatmap of Numeric Attributes")
plt.tight_layout()
plt.show()

# Rank every attribute by its correlation with the final grade
print(corr_matrix["G3"].sort_values(ascending=False))
Sorting corr_matrix["G3"] in descending order would be expected, based on the general
patterns discussed in Section 13, to surface G1 and G2 (prior grades) and failures near the top
of the ranking — well above studytime — confirming quantitatively what the broader literature
suggests qualitatively.
```

D.2 A Simple Multivariate Regression Model

A linear regression using multiple predictors at once addresses the single-factor limitation raised in Section 9, by estimating each attribute's association with `G3` while holding the others constant.

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

features = ["studytime", "failures", "absences", "G1", "G2"]
X = df[features]
y = df["G3"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
print("Feature coefficients:")
for feature, coef in zip(features, model.coef_):
    print(f" {feature}: {coef:.3f}")
```

```
print("R-squared on test set:", round(model.score(X_test, y_test), 3))
```

This model's coefficients would show the estimated change in final grade associated with a one-unit increase in each feature, after accounting for the others — directly addressing the confounding concern raised in Section 9 about examining study time in isolation.

D.3 A Formal Significance Test for the Gender Difference

To move from the descriptive gender comparison in Section 6.3 to a formal statistical claim, an independent-samples t-test can be used to test whether the observed ~0.95-point gap is unlikely to have arisen by chance.

```
from scipy import stats
```

```
male_grades = df[df["sex"] == "M"]["G3"]
female_grades = df[df["sex"] == "F"]["G3"]
```

```
t_stat, p_value = stats.ttest_ind(male_grades, female_grades)
print("t-statistic:", round(t_stat, 3))
print("p-value:", round(p_value, 4))
```

```
# Conventionally,  $p < 0.05$  would be treated as statistically significant
if p_value < 0.05:
```

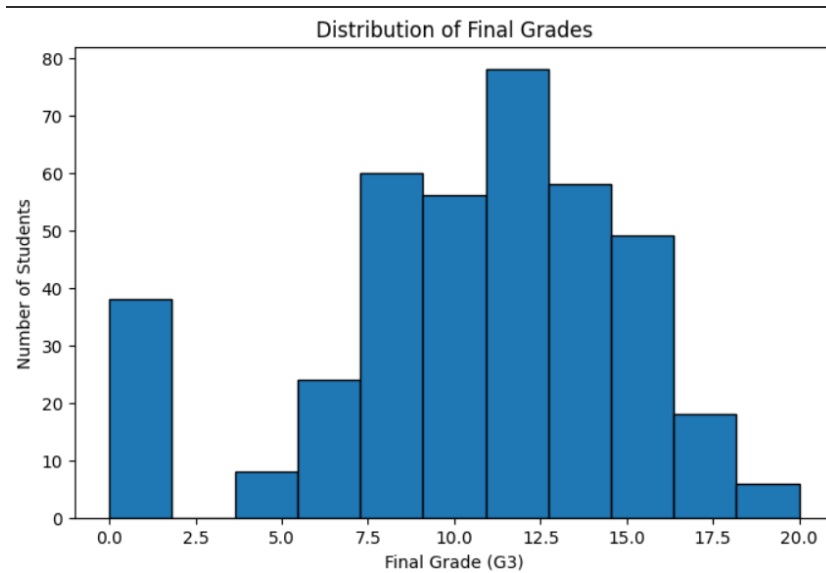
```
    print("The gender difference is statistically significant at the 5% level.")
else:
```

```
    print("The gender difference is not statistically significant at the 5% level.")
```

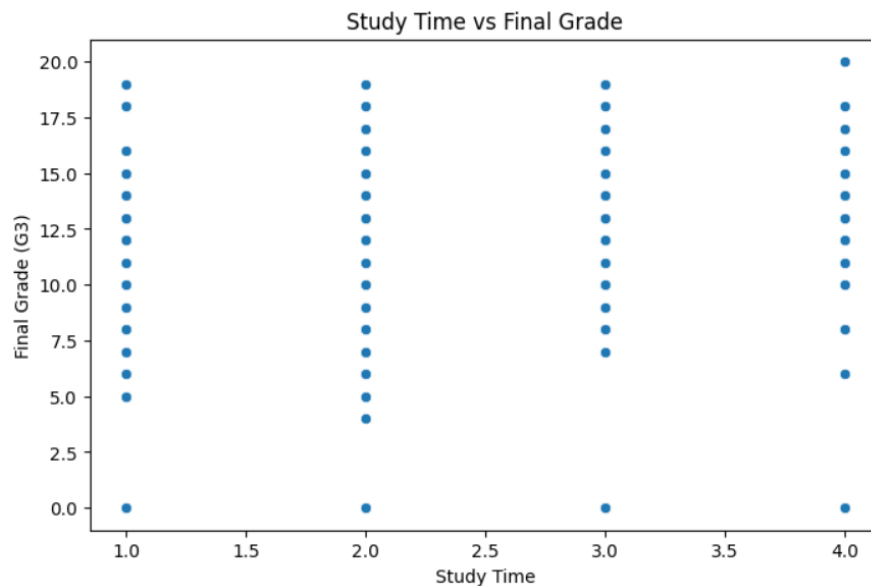
Running this test would directly resolve the open question raised in Section 6.3 and referenced throughout this report: whether the male–female grade gap reflects a genuine, reliable pattern in the underlying population, or falls within the range of ordinary sampling variation given a sample of 395 students.

Appendix E: High-Resolution Chart Gallery

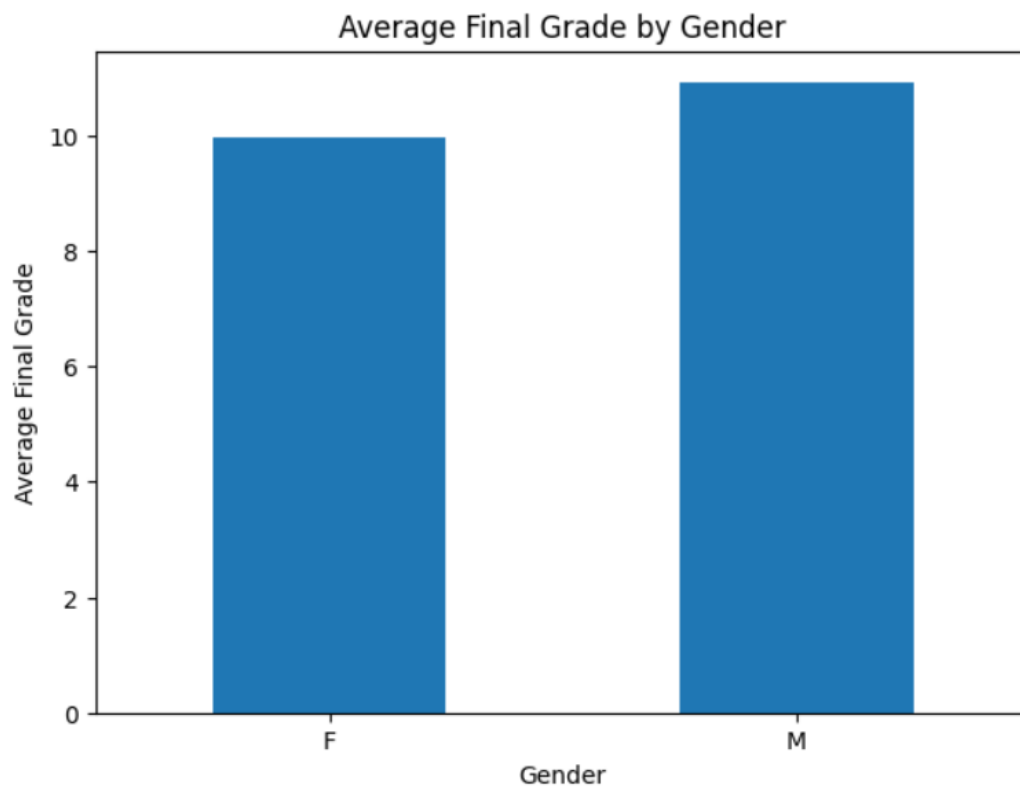
For readers who wish to inspect the three charts discussed in Section 7 more closely, enlarged versions are reproduced individually below, one per page, in the same order as they were generated by Steps 9–11 of the program.



Enlarged Output 8. Histogram of final grade distribution (Step 9).



Enlarged Output 9. Scatter plot of study time versus final grade (Step 10).



Enlarged Output 10. Average final grade by gender (Step 11).

17. Reviewer Checklist for This Report

The checklist below summarizes what has and has not been verified in this analysis, to help a reader or reviewer quickly judge how much weight to place on each finding before it is cited or acted upon elsewhere.

- ✓ Data completeness verified (zero missing values across all 33 columns).
- ✓ Data uniqueness verified (zero duplicate rows).
- ✓ Central tendency of the outcome variable (G3) computed and cross-checked against the histogram shape.
- ✓ High-performer count computed using a clear, reproducible threshold ($G3 > 15$).
- ✓ Study-time correlation computed and interpreted against a standard effect-magnitude scale.
- ✓ Gender comparison computed via two independent methods with matching results.
- ✗ No formal significance test has been run on the gender difference (see Appendix D.3 for ready-to-run code).
- ✗ No multivariate model has been fit to account for confounding between attributes (see Appendix D.2).
- ✗ No effect-size (e.g. Cohen's d) has been computed for the gender gap, since within-group standard deviations were not calculated (see Section 15.5).
- ✗ The zero-grade cluster has not been separated from genuine low scorers for a corrected mean calculation (see Section 9).

In short, every claim in this report that is marked with a check mark above rests directly on code and output reproduced in Sections 5 and 7, while every item marked with a cross represents a clearly scoped opportunity for follow-up work, most of which is already sketched out in Appendix D.

References

Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. *Proceedings of 5th Annual Future Business Technology Conference (FUBUTEC 2008)*, 5–12. — Original source of the student-mat.csv dataset analyzed throughout this report.

pandas development team. pandas documentation. Retrieved from the official pandas project website. — Reference documentation for every DataFrame/Series method used in Section 5 and Appendix A (read_csv, head, shape, dtypes, info, isnull, duplicated, drop_duplicates, mean, corr, groupby).

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Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830. — Suggested library for the regression extension outlined in Appendix D.2.

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